Scaling Automation

Execution Environments & Configuration as Code

A look at how our GitHub Actions pipeline declaratively manages the **Ansible Automation Platform (AAP)** using portable, versioned Execution Environments.

Agenda

1. What is an Execution Environment (EE)?
2. Why EEs matter in our workflow
3. Anatomy of our EE
4. CI/CD pipeline — from commit to AAP
5. Deep dive: `deploy_cac.yml`
6. Business benefits
7. Q&A

What is an Execution Environment?

A **container image** that serves as the control node for Ansible automation.

Components inside the image:

- Base OS — *RHEL 9 Minimal*
- `ansible-core` & `ansible-runner`
- Python & system dependencies — `gcc`, `python3.12-devel`, …
- Ansible Collections — `ansible.controller`, `ansible.platform`, …

> **Purpose:** Eliminates the *"it works on my machine"* problem by packaging every automation dependency into one portable container.

Why EEs Matter in Our Workflow

Consistency

Same dependencies in **local dev**, **GitHub Actions CI/CD**, and the **AAP cluster**.

Security & Control

We explicitly declare every collection and package.
Built reproducibly via `ansible-builder`.

Decoupling

The AAP **control plane** is decoupled from the **execution plane**.
EEs can be updated independently of the underlying AAP infrastructure.

Anatomy of Our Execution Environment

Source of truth: `execution-environment/execution-environment.yml`
Base image: `ee-minimal-rhel9:latest` *(AAP 2.6)*

Dependencies installed at build time:

- Python: `ansible-core`, `ansible-runner`, `setuptools`, `wheel`
- Collections from `requirements.yml`
  — pulled from **Red Hat Automation Hub** and **Galaxy**

Build tool: `ansible-builder` compiles this YAML into a standard OCI image.

How This Code Pushes to AAP

The CI/CD Pipeline — `.github/workflows/deploy.yml`

Step	Stage	What happens
1	**Lint & Validate**	`yamllint` + `ansible-lint` enforce standards
2	**Build & Publish EE**	Build image, push to **GHCR** and **Private Automation Hub**
3	**Dry-Run Validation**	Run `--syntax-check` inside the new EE against Dev
4	**Declarative Deploy**	Execute `deploy_cac.yml` inside the EE → AAP Controller API

Deep Dive: `deploy_cac.yml`

Declarative: We define *what* AAP should look like — no UI clicks.

Phased execution:

- **Phases 1–3** — Load baseline (`configs/all`) + env overrides (Dev / Prod)
- **Phase 4** — Apply to AAP in strict dependency order:
    i. Organizations
    ii. Execution Environments
    iii. Projects & Inventories
    iv. Hosts & Groups
    v. Job Templates & Workflows
    vi. RBAC / Roles

Business Benefits

Auditability

Every AAP change is a **Git commit** with PR review and full history.

Disaster Recovery

Lost a cluster? Rebuild the entire AAP configuration **from this repo** in minutes.

Velocity

Teams propose changes via **Pull Requests**, not IT tickets.

Questions?

EEs · `ansible-builder` · `deploy_cac.yml`

Thanks for your time.